

TECHNO INDIA COLLEGE OF TECHNOLOGY

Department of Computer Science & Engineering

Computer Network Lab Manual

Paper Code: CS692



Objective: Explore the concept of de-multiplexing at transport layer

LAB 8

Demonstration of multiplexing & de-multiplexing of messages using Wireshark

3
Hours

Experiment 1:

Goal: Ms. X finds ssh protocols fascinating and now wants to capture what happens when two ssh sessions are established (at about the same time) between her machine and the same remote host. Since the sessions have same source and destination IP address, she wants to figure out how the sessions are uniquely identified. Help her capture such a trace. There should be no other background traffic.

Guidance:

1. Use two windows to run each ssh session, type the command in both windows and run these commands as close in time as possible. (ssh to a neighbouring machine)
2. When opening the trace in wireshark, we will use filters to filter out unnecessary packets and capture just the first packet of both sessions in either direction (client to server and back). Type *tcp.flags.syn == 0x02* in the filter's field (this essentially makes use of the *syn* flag of TCP header, which is set to 1 only in the first packet of the TCP connection). You should see 4 packets listed.

Report:

1. Explain your design by specifying the exact commands (with options) you will run and in which order. Avoid description unless absolutely necessary.
2. What is the port number used by the remote machine for the first and the second ssh session? Are both sessions connected to the same port number on the remote machine? How do you think the ssh application at remote machine distinguishes between the two sessions?
3. When your machine receives packets from the remote host, how does the TCP layer figure out to which ssh session this packet has to be passed? Specify the value of the fields used by TCP to do this.
4. Name your trace file as "exercise1.out" and add the file to your roll-number directory.